

CivicLens RAG - NYC 311 Operations Copilot

Grounded NYC 311 documentation Q&A with hybrid retrieval, validated citations, explicit abstention handling, and bounded analytics.

APPROVED LOCAL EVALUATION

Real-local profile; not hosted runtime metrics

83.9%

Recall@5 (n=14)

92.9%

Expected-source retrieval (n=14)

95.8%

Application routing accuracy (n=24)

BUSINESS / PRODUCT CONTEXT

1 Problem

Operational documentation and runbooks are difficult to search reliably while preserving source traceability.

2 Solution

Combine semantic and full-text retrieval, fuse results with RRF, and validate citations before returning answers.

3 Outcome

A hosted portfolio application with measurable retrieval gains, cited answers, and visible failure modes.

SYSTEM ARCHITECTURE

One application boundary, two deliberately separate answer paths

The browser delegates AI behavior to FastAPI; CivicLens routes to document RAG or bounded analytics.



PATH A

Hybrid RAG | PostgreSQL + pgvector

Neon in the hosted demo. Semantic + full-text retrieval + RRF -> grounded answer + validated citations.

PATH B

Bounded Analytics | checked-in sample CSVs

Exactly four typed, allowlisted, read-only tools; no unrestricted text-to-SQL.

KEY TAKEAWAYS

- Hybrid retrieval raised Recall@5 from 66.1% to 83.9% on the approved retrieval subset.
- Expected-source retrieval improved from 78.6% to 92.9%.
- Known abstention and routing failures remain visible in the evaluation record.

What the evaluation showed

The approved real-local comparison is a small portfolio benchmark. It measures retrieval behavior and application failure modes, not production reliability.

RETRIEVAL COMPARISON - N=14

Strategy	Recall@5	MRR	Expected source
Semantic	66.1%	58.6%	78.6%
Hybrid	83.9%	70.7%	92.9%
Hybrid + reranking	82.1%	76.2%	92.9%

KNOWN WEAKNESS

33.3%

Safe no-answer accuracy (2/6)

Four questions expected to abstain were answered. One adversarial question routed to predefined analytics and was counted as unsupported.

INTERPRETATION

Hybrid search improved retrieval recall; reranking improved ranking quality.

Reranking raised MRR from 70.7% to 76.2%, but Recall@5 moved from 83.9% to 82.1%. The result is a trade-off, not a universally superior strategy.

CONTROLLED OUTPUTS

Application-owned safeguards keep outputs inspectable

● Grounding & citations

Retrieved evidence feeds generation; CivicLens validates stable citations before response.

● Bounded analytics

Four typed, allowlisted tools read checked-in sample CSVs; no arbitrary SQL or code execution.

● Hosted proof

Vercel Next.js calls Render FastAPI; document RAG uses Neon-hosted PostgreSQL + pgvector.

SCOPE & LIMITATIONS

- Curated, version-controlled corpus; no live NYC 311 ingestion.
- 24-question fixture (14 retrieval-eligible); not a production benchmark.
- Free-tier hosted demo may cold-start; no production authentication/authorization, HA, autoscaling, or SLA.
- Four read-only sample-data tools; no unrestricted production text-to-SQL.

PORTFOLIO EVIDENCE

[Live Demo](#) [Case Study](#) [Repository](#) [Evaluation](#)